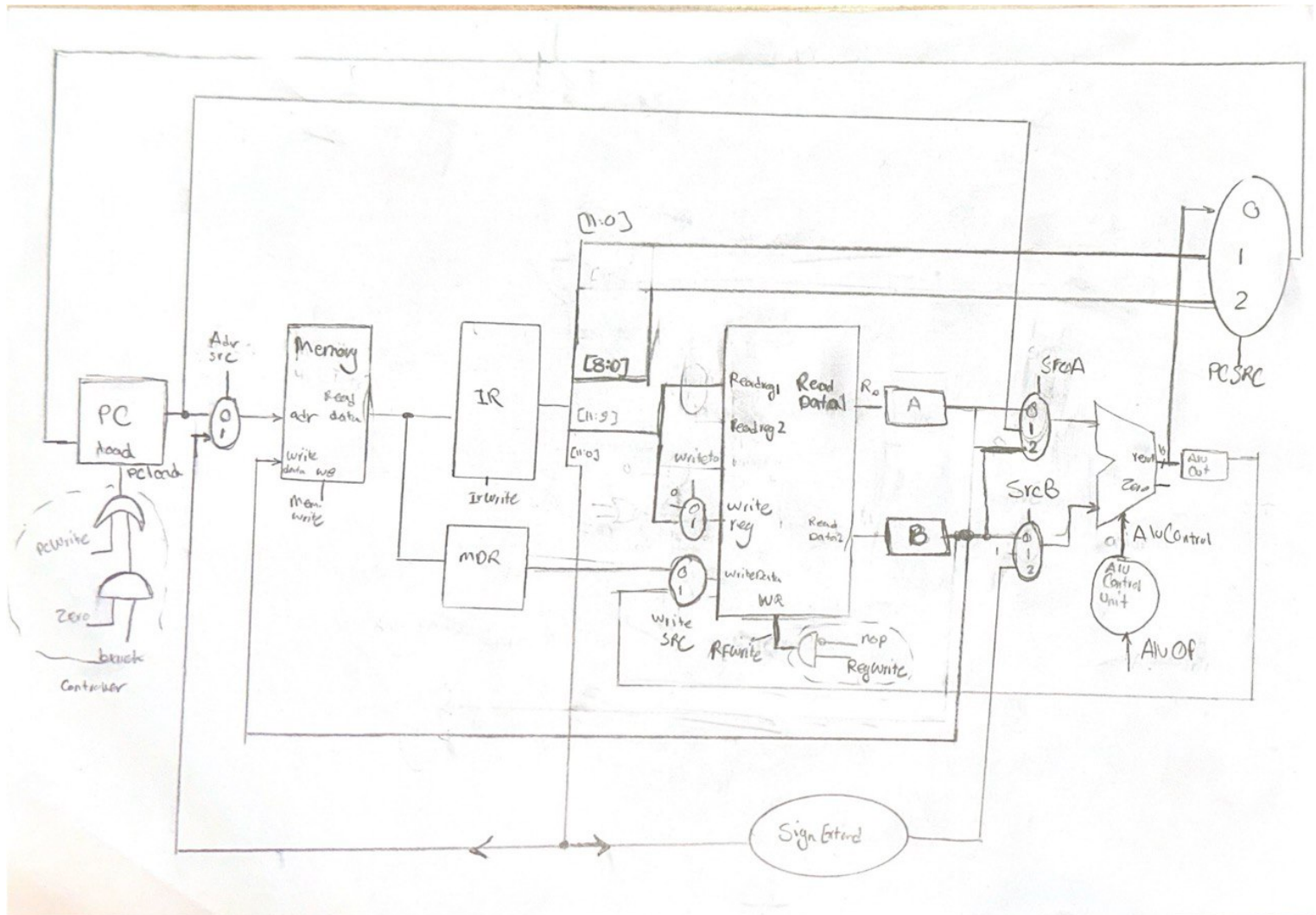
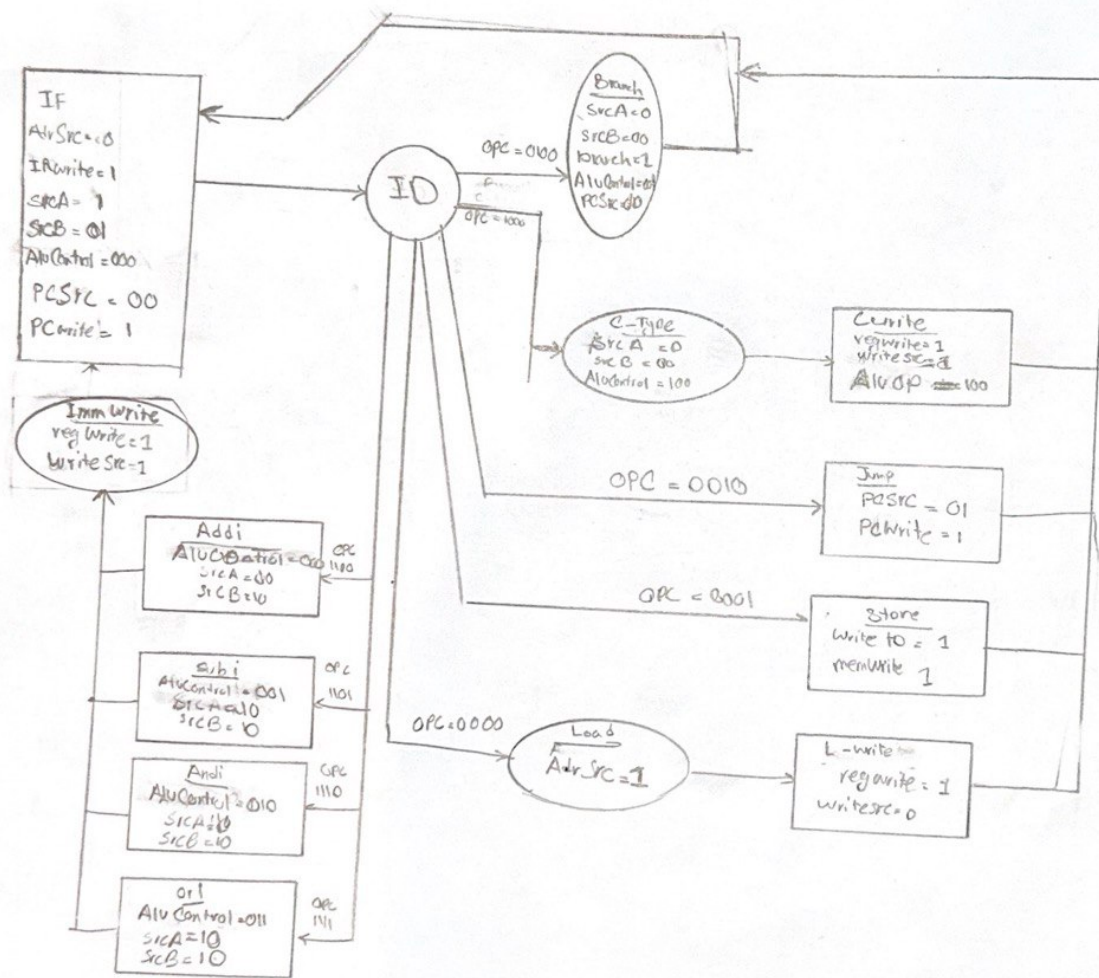


Datapath



Controller



State	PCload	AdrSrc	MemWrite	IRWrite	RFWrite	WriteTo	WriteSrc	SrcA	SrcB	AluOP	PCSrc	branch
IF	1	0	0	1	0	-	-	01	01	000	00	0
ID	0	-	0	0	0	-	-	-	-	-	-	0
Branch	-	-	0	0	0	-	-	00	00	001	10	1
C-Type	0	-	0	0	0	-	-	00	00	100	-	0
C-Write	0	-	0	0	1	0	1	-	-	100	-	0
Jump	1	-	0	0	0	-	-	-	-	-	01	0
Store	0	1	1	0	0	-	-	-	-	-	-	0
Load	0	1	0	0	0	-	-	-	-	-	-	0
L-Write	0	-	0	0	1	0	0	-	-	-	-	0
Addi	0	-	0	0	0	-	-	10	10	000	-	0
Subi	0	-	0	0	0	-	-	10	10	001	-	0
Andi	0	-	0	0	0	-	-	10	10	010	-	0
Ori	0	-	0	0	0	-	-	10	10	011	-	0
ImmWrite	0	-	0	0	1	0	1	-	-	-	-	0

Output

```
ModelSim> run -all
# Sum :      55
# ** Note: $stop      : D:/University/5th Semester/Digital Systems II/Computer Assignments/CA3/Testbench.sv(14)
#   Time: 20020 ns   Iteration: 0   Instance: /MultiCycle_TB
# Break in Module MultiCycle_TB at D:/University/5th Semester/Digital Systems II/Computer Assignments/CA3/Testbench.sv line 14
```

Console